

The Evolution of AI Agents and Modern AI Engineering

A Technical Research Report

1. LLM Applications and the Rise of AI Agents

LLMs such as GPT-4, Claude, and Gemini started as text-completion engines inside reactive LLM applications — chatbots, summarizers, and RAG pipelines that answer once and stop. Between 2023 and 2026 the field shifted toward AI agents: LLM-driven systems that autonomously plan, call tools, remember context, and take multi-step actions toward a goal (Masterman et al., 2024).

2. AI Agents: Core Architecture

An AI agent perceives input, reasons about a goal, selects actions (including tool calls), and acts on an environment, iterating on feedback until the goal is met. Surveys (Xi et al., 2025; MedRAX, 2025) converge on three core components — a reasoning engine (the LLM), perception modules, and action mechanisms — wrapped by two more subsystems: memory and planning.

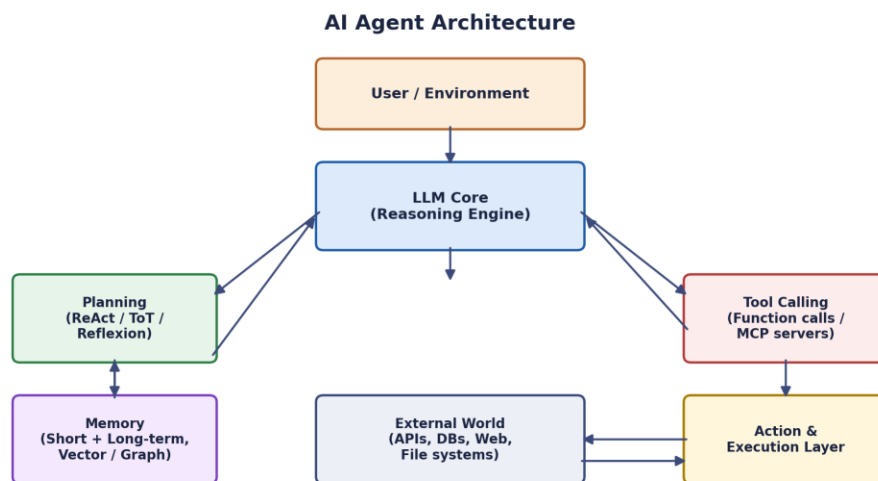


Figure 1: AI agent architecture — LLM core linked to planning, memory, tool-calling and action layers in a perceive-reason-act loop.

3. Tool Calling and the Model Context Protocol (MCP)

Tool calling lets an LLM invoke external functions instead of relying only on its parametric knowledge. Anthropic's Model Context Protocol (MCP, late 2024) standardized this via a client-server model: an MCP host connects through an MCP client to independent MCP servers exposing tools over JSON-RPC (SUMO-MCP, 2025). By 2025-2026 OpenAI, Google, Cursor, and Cline had adopted MCP, and in December 2025 Anthropic donated it to the Agentic AI Foundation (Linux Foundation) with OpenAI and Block.

4. Memory: Giving Agents Persistence

Since context windows are finite, agents need external memory. MemGPT (Packer et al., 2023/24) pioneered an OS-style paging approach; MemoryBank (Zhong, 2024) and Zep (Rasmussen, 2025, temporal knowledge graphs) extend long-term recall; Mem0 (Chhikara, 2025) and A-Mem (Xu, 2025) add incremental, self-organizing memory. Systems typically split memory into short-term, episodic, and long-term/semantic stores retrieved via vector or graph search.

5. Planning: Reasoning Before and During Action

6.

Chain-of-Thought (Wei et al., 2022) elicits step-by-step reasoning with no environment interaction. ReAct (Yao et al., 2023) interleaves thought-action-observation loops. Tree-/Graph-of-Thoughts (Yao 2023; Besta 2024) add branching search with self-evaluation, and Reflexion (Shinn et al., 2023) stores verbal self-critiques across episodes for cross-trial learning. 2026 research shows local step-by-step reasoning alone still struggles on long-horizon tasks, motivating explicit search and hierarchical planning.

7. Multi-Agent Systems

Complex work is increasingly split across cooperating agents. MetaGPT (Hong et al., ICLR 2024) encodes SOPs into role-based agent pipelines; AutoGen/AG2 (Microsoft) uses conversational, peer-to-peer coordination; CrewAI and CAMEL add role-based teams. Google's Agent2Agent (A2A) protocol (2025) standardizes task delegation between agents from different vendors, complementing MCP's tool-access layer. A 2025 LangChain survey found 57.3% of surveyed organizations already run agents in production.

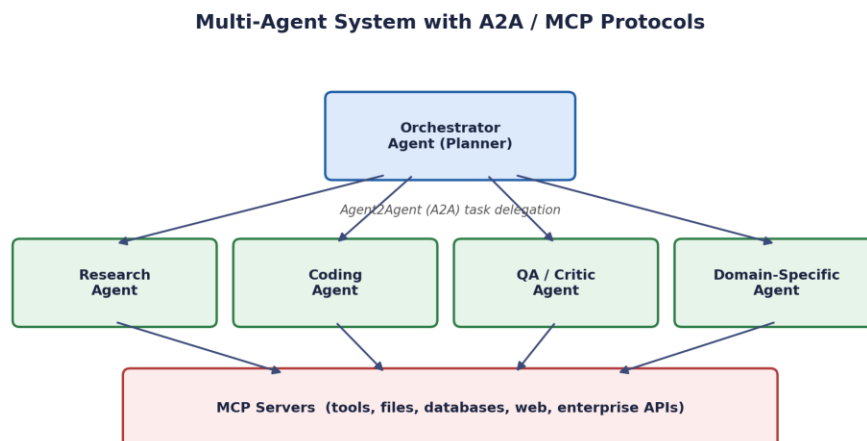


Figure 2: Multi-agent system — orchestrator delegates to sub-agents over A2A; each reaches shared tools via MCP servers.

8. Real-World Examples

- **Coding:** Claude Code / Claude Agent SDK plan multi-step coding tasks, call MCP tools, and coordinate parallel agent teams.

- **Software engineering:** SWE-agent and Devin-style systems resolve GitHub issues autonomously in a ReAct-style loop.
- **Healthcare:** MedRAX (2025) pairs an LLM core with medical-imaging tools to interpret chest X-rays via agentic tool calls.
- **Enterprise:** Enterprises connect CRMs, ticketing, and data warehouses to agents via MCP servers, avoiding custom integrations.

9. Future of AI Engineering

The field is consolidating around MCP (tool access), A2A (agent delegation), and conventions like AGENTS.md. Expect growth in interoperable protocols, hybrid vector/graph/episodic memory, formal agent-reliability benchmarks (MCPToolBench++, LongMemEval), and stronger governance — Gartner projects 40% enterprise penetration by end-2026 alongside a wave of project cancellations by 2027 from cost and reliability gaps. AI engineering is shifting from prompt engineering to full agentic system design.

References

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